

The Projection Principle:

**A First-Principles Theory of Bounded Receipted Execution
for Systems Beyond Human Cognitive Load**

Derived and formalized in the Praxis / Capability Physics program

Genesis, Day One

Abstract

We develop, from first principles, a theory for trusting computational systems whose internal state and control flow exceed any human observer’s working-memory capacity. The central object is the *Chatman equation* $\mathcal{A} = \mu(\mathcal{O}^*)$, read as: an artifact is the lawful image, under a deterministic manufacturing morphism μ , of an *admitted* observation. We formalize admission as a retraction onto a decidable sub-language (the *Rice quarantine*), define a *receipt* as a projection π from an arbitrarily high-dimensional execution trace onto a bounded coordinate, and prove three results: (i) the *Faithful Projection Theorem*, that a collision-resistant chained receipt lets a bounded verifier reject any perturbed execution with overwhelming probability while reading only $O(1)$ symbols; (ii) the *Comprehension–Verification Gap*, a quantitative statement that verification cost is bounded by the projection dimension and is independent of the interior dimension; and (iii) a *Conservation Law of Consequence*, that under the admission discipline every actuated effect is the image of exactly one admitted observation and carries a unique receipt. We give the algebra of the admission monoid, the calculus of the cost/attention fluent that governs scheduling, and the geometry of the token-flow state space in which conformance is a distance-to-a-polytope. The thesis closes by exhibiting a concrete instance — a seven-day autonomous build that spent on the order of 2.5×10^6 machine-generated tokens no human read — and showing that its trustworthiness rests entirely on boundary receipts sized for a human, not on comprehension of the interior.

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Chapter 1

Introduction: The Standard Was Never Comprehension

A persistent assumption in engineering practice holds that a trustworthy system must be one a competent human can hold in mind. This assumption is false for the systems that matter most, and the falsity is not incidental but structural. A branchless SIMD document parser classifies sixty-four input bytes per vector instruction through carry-less arithmetic that no engineer traces mentally at runtime; a triple store answers a query over 10^{11} facts by walking permutation indexes whose join plan no engineer reconstructs by hand. Both are trusted universally. The trust does not arise from comprehension. It arises from *differential agreement, adversarial fuzzing, invariants, and checked projections of the output*.

This thesis takes that observed practice and derives it as a theorem. We fix, once, the quantity that makes the problem well posed: human working memory admits a small constant number of simultaneously manipulable chunks, which we write κ and take, following the cognitive-load literature, to be a fixed small integer (canonically $\kappa \approx 4$). A system whose faithful description requires more than κ chunks cannot be verified by direct comprehension. We do not respond by shrinking the system. We respond by inserting, between the system and the human, a map whose codomain has dimension at most κ and whose fidelity is guaranteed by the mechanism rather than by the reader. That map is the receipt, and this thesis is its theory.

Contributions and structure. Chapter 2 axiomatizes observation, admission, and the Rice quarantine, and constructs the admission monoid (algebra). Chapter 3 defines the manufacturing morphism μ and the receipt projection π , and proves the Faithful Projection Theorem and the Conservation Law. Chapter 4 develops the calculus of the attention fluent and derives lexicographic cost arbitration as a limit of a weighted objective. Chapter 5 gives the geometry: the marking polytope, conformance as distance, and separability as a decomposition of that geometry. Chapter 6 instantiates the theory on the 2.5-million-token build and states the Comprehension–Verification Gap as a corollary with measured constants.

Chapter 2

Observation, Admission, and the Algebra of the Quarantine

2.1 First principles

We begin with the least structure that makes “lawful action” expressible.

Axiom 2.1 (Observation). There is a set $\mathcal{O}$, the *observation space*, whose elements are arbitrary finite records: logs, model outputs, sensor frames, human claims, tool responses. $\mathcal{O}$ carries no decidable semantics: the map “does this observation mean what it purports to mean” is not assumed computable.

Axiom 2.1 is a modeling commitment, not a limitation, and it is forced by a classical result.

Theorem 2.1 (Rice, specialized). *Let P be any non-trivial semantic property of the programs (equivalently, of the meanings) that observations in $\mathcal{O}$ may encode. Then $\{o \in \mathcal{O} : P(o)\}$ is undecidable.*

Proof sketch. Rice’s theorem states that every non-trivial property of the language recognized by a Turing machine is undecidable. Observations are unrestricted finite encodings and may encode arbitrary machines; any non-trivial semantic predicate on them therefore inherits undecidability by the standard reduction from the halting problem. $\square$

Theorem 2.1 is the reason the whole apparatus exists: one cannot admit an observation by deciding its semantics. One admits it by *retracting* it onto a sub-language on which the properties of interest *are* decidable.

Definition 2.1 (Admitted space and admission). Fix a decidable sub-collection $\mathcal{O}^* \subseteq \mathcal{O}$ — the *admitted space* — on which a finite battery of obligations $g_1, \dots, g_m : \mathcal{O} \rightarrow \{0, 1\}$ is computable. The *admission map* is the partial retraction

$$\alpha : \mathcal{O} \rightarrow \mathcal{O}^* \cup \{\perp\}, \quad \alpha(o) = \begin{cases} \rho(o) \in \mathcal{O}^* & \text{if } \bigwedge_{i=1}^m g_i(o) = 1, \\ \perp & \text{otherwise,} \end{cases}$$

where $\rho : \mathcal{O} \rightarrow \mathcal{O}^*$ is a computable bounding/normalization and $\perp$ is the distinguished *refusal*. Admission is idempotent on its image: $\alpha \circ \alpha = \alpha$ (a retraction).

Remark. $\perp$ is a first-class value, not an exception. A refusal is a lawful output of a correctly functioning admission: the system computed that an obligation failed and recorded which one. This is the formal content of “refusal is data.”

2.2 The admission monoid

Obligation checking composes, and the composition has algebraic structure we will use in the geometry chapter.

Construction 2.1 (Denial algebra). Let $D = (\{0, 1\}^n, \vee, \mathbf{0})$ be the commutative idempotent monoid of *denial words*: n independent lanes, componentwise OR, identity $\mathbf{0}$ (“admitted, all lanes clear”). Each obligation g_i contributes a lane map $d_i : \mathcal{O} \rightarrow \{0, 1\}^n$ with $d_i(o) = \mathbf{0} \iff g_i(o) = 1$. The total denial of an observation is

$$d(o) = \bigvee_{i=1}^m d_i(o) \in \{0, 1\}^n, \quad \alpha(o) \neq \perp \iff d(o) = \mathbf{0}.$$

Proposition 2.2 (D is a bounded semilattice). *($\{0, 1\}^n, \vee$) is a commutative idempotent monoid; equivalently the join-semilattice of the Boolean lattice $2^{[n]}$ with bottom $\mathbf{0}$. Consequently admission is monotone: adding obligations can only enlarge the denial (turn admits into refusals), never the reverse.*

Proof. Idempotence $x \vee x = x$, commutativity, and associativity are the defining laws of $\vee$ on each coordinate; the product of semilattices is a semilattice. Monotonicity: if $G' \supseteq G$ are obligation sets then $d_{G'}(o) = d_G(o) \vee \bigvee_{i \in G' \setminus G} d_i(o) \succeq d_G(o)$ in the coordinatewise order, and $d(o) = \mathbf{0}$ is the unique minimum, so $\{o : d_{G'}(o) = \mathbf{0}\} \subseteq \{o : d_G(o) = \mathbf{0}\}$. $\square$

Proposition 2.2 is the algebraic form of “the quarantine only ever tightens.” It also gives the refusal taxonomy its type: a refusal *category* is the join-irreducible support $\text{supp } d(o) \subseteq [n]$, and the eight-bucket taxonomy of the implementation is a fixed surjection $[n] \twoheadrightarrow \{1, \dots, 8\}$ onto named categories.

Chapter 3

Manufacture and the Receipt Projection

3.1 The Chatman equation

Definition 3.1 (Manufacturing morphism). A *manufacturing morphism* is a deterministic computable map $\mu : \mathcal{O}^* \rightarrow \mathcal{A}$ from admitted observations to artifacts, subject to two laws:

- (M1) **Determinism / reproducibility:** $\mu(x)$ depends only on x ; equal admitted inputs yield bit-identical artifacts.
- (M2) **Boundedness:** μ factors through a representation of size bounded by fixed structural constants (arity, depth, conjunct count ≤ 8 in the canonical profile), so that μ terminates and its cost is a priori bounded.

The governing identity, the *Chatman equation*, is

$$\mathcal{A} = \mu(\mathcal{O}^*), \quad \text{operationally} \quad a = \mu(\alpha(o)) \text{ whenever } \alpha(o) \neq \perp. \quad (3.1)$$

Nothing is manufactured from a raw observation; only from an admitted one. Equivalently, μ is undefined on $\mathcal{O} \setminus \mathcal{O}^*$ by construction, and $\perp$ propagates: $\mu(\perp) = \perp$.

Proposition 3.1 (Functoriality of manufacture). *If admitted observations form a category $\mathbf{Adm}$ whose morphisms are the lawful edits $\delta : x \rightarrow x \sqcup \delta$ (join with an admitted increment), and artifacts form $\mathbf{Art}$ under composition of edits, then a deterministic μ with $\mu(x \sqcup \delta) = \mu(x) * \mu_\delta$ for a partial action $*$ is a functor $\mathbf{Adm} \rightarrow \mathbf{Art}$. Composition of manufactures is manufacture of composition: $\mu_2 \circ \mu_1$ manufactures the composite artifact.*

Proof. Identity edit maps to identity artifact by (M1); associativity of $*$ follows from determinism applied to the associative $\sqcup$ on admitted increments. Hence the assignment on objects and edits preserves identities and composition, i.e. is a functor. $\square$

3.2 Receipts as projections

Let an *execution trace* be the full internal record of a manufacture: every intermediate marking, branch, token movement, and sub-artifact. Traces live in a space Σ of astronomically high —

indeed unbounded — dimension. A human verifier has capacity κ . The receipt is the map that reconciles the two.

Definition 3.2 (Receipt projection). Fix a collision-resistant hash $H : \{0, 1\}^* \rightarrow \{0, 1\}^{256}$. For a trace $\sigma \in \Sigma$ with payload bytes $b(\sigma)$, previous chain value h_- , and metadata θ (instruction index, activity, timestamp, denial word), the *frame* is a fixed-width encoding $\text{fr}(\sigma) = \langle \theta, H(b(\sigma)) \rangle$ and the *chained receipt* is

$$h_+ = H(h_- \parallel \text{fr}(\sigma)). \quad (3.2)$$

The *receipt projection* is

$$\pi : \Sigma \longrightarrow \mathcal{R}, \quad \pi(\sigma) = (\text{verdict}, h_+, \varphi(\sigma), \text{reason}),$$

a tuple of at most κ chunks: a Boolean verdict, the 256-bit chain value, one scalar metric φ , and one refusal reason (empty on accept).

Theorem 3.2 (Faithful Projection). *Let σ be an admitted execution and h_+ its chained receipt (3.2). Suppose an adversary presents a perturbed trace $\sigma' \neq \sigma$ agreeing with σ on the claimed receipt h_+ . Then producing such a σ' requires exhibiting a collision of H ; hence under the collision resistance of H , a verifier who recomputes (3.2) from σ' and compares 256 bits rejects σ' except with probability negligible in the security parameter. Moreover the verifier reads $O(1)$ symbols — exactly the κ -chunk projection — independent of $\dim \Sigma$.*

Proof. The chain (3.2) commits to $b(\sigma)$ through $H(b(\sigma))$ inside fr , and to all predecessors through h_- , which itself commits recursively. Thus h_+ is a function of the entire causal history. If $\sigma' \neq \sigma$ but the recomputed chain equals the claimed h_+ , then either some $H(b(\sigma')) = H(b(\sigma))$ with $b(\sigma') \neq b(\sigma)$ (a collision of H), or the recomputation differs and the 256-bit comparison fails. By collision resistance the first event has negligible probability. The verifier’s work is one hash recomputation per frame it chooses to check and one 256-bit equality; to *reject* it suffices to find the first frame whose recomputation disagrees, and to *accept* the projection it reads only $(\text{verdict}, h_+, \varphi, \text{reason})$, a quantity of size independent of $\dim \Sigma$. $\square$

Corollary 3.3 (Comprehension is unnecessary). *Trust in an accepted artifact is a function of $\pi(\sigma)$ and the soundness of the verifier, not of any human’s comprehension of σ . Formally, the verifier’s accept predicate factors as $\text{Verify} = V \circ \pi$ for a fixed $V : \mathcal{R} \rightarrow \{0, 1\}$ of input size $\leq \kappa$.*

Theorem 3.4 (Conservation of Consequence). *Under Definitions 2.1, 3.1, 3.2, every actuated artifact $a \in \text{im } \mu$ is the image of exactly one admitted observation and is committed by exactly one receipt chain position; and no $a \notin \text{im } \mu$ is actuated. Consequently the map $a \mapsto h_+(a)$ from actuated artifacts to receipt positions is injective up to hash collision.*

Proof. Actuation is gated: by (3.1), a is actuated only if $a = \mu(x)$ for some $x = \alpha(o) \neq \perp$; hence $\text{im } \mu$ exhausts the actuated set and its complement is never actuated. Determinism (M1) makes $x \mapsto \mu(x)$ a function, and the chain appends exactly one position per manufacture, so each actuation occupies a unique chain index. Injectivity up to collision is Theorem 3.2 applied to the frames committing $b(\mu(x))$. $\square$

Theorem 3.4 is the “conservation law”: consequence is neither created without an admitted cause nor destroyed without a receipt, and the chain preserves order. It is the discrete analogue of a continuity equation, made precise in the next chapter.

Chapter 4

The Calculus of Attention and Cost

4.1 The attention fluent

Scheduling admitted actions consumes a shared, conserved resource we call the *attention fluent* $\varphi_a(t)$ — the operator/agent capacity available at time t . Borrowing the term *fluent* in its original Newtonian sense (a quantity that flows), we model it as a continuous state variable obeying a balance law.

Definition 4.1 (Attention balance). Let $c(t) \geq 0$ be attention capacity and let admitted action j draw rate $r_j \geq 0$ while active on an interval $[s_j, e_j]$. The instantaneous free capacity is

$$f(t) = c(t) - \sum_{j: t \in [s_j, e_j]} r_j.$$

A schedule is *feasible* iff $f(t) \geq 0$ for all t , i.e. the conservation constraint $\sum_j r_j \mathbf{1}_{[s_j, e_j]}(t) \leq c(t)$ holds pointwise.

Proposition 4.1 (Makespan as a functional). *For a fixed feasible schedule the makespan $T = \max_j e_j$ is the terminal time of the capacity balance, and the total attention consumed is the integral*

$$\int_0^T \left(\sum_j r_j \mathbf{1}_{[s_j, e_j]}(t) \right) dt = \sum_j r_j (e_j - s_j),$$

a conserved quantity independent of the schedule's ordering. Thus reordering redistributes when attention is spent but not how much; the optimization target is the makespan functional $T[\cdot]$, minimized subject to precedence and the pointwise capacity constraint.

Proof. Linearity of the integral gives the second identity; the first is the definition of makespan. Independence of ordering follows because each term $r_j(e_j - s_j)$ depends only on action j 's duration and rate, not on its placement. $\square$

4.2 Lexicographic arbitration as a limit

When several admitted plans compete, the implementation orders them by a *cost vector* with lawfulness dominating. We derive this lexicographic order as a limit of ordinary weighted optimization, which explains why “admitted” is not merely a large weight but an infinite one.

Definition 4.2 (Cost vector). To a candidate plan assign $\mathbf{c} = (c_0, c_1, \dots, c_k)$ with $c_0 \in \{0, 1\}$ the *unadmitted* indicator ($0 = \text{admitted}$) and $c_1, \dots, c_k$ the ordered secondary costs (mutation risk, attention-seconds, tokens, latency, switches). Order plans by the lexicographic order $\preceq_{\text{lex}}$ on $\mathbf{c}$.

Theorem 4.2 (Lexicographic order is the weight limit). *Let $C_\lambda(\mathbf{c}) = \sum_{i=0}^k \lambda^{k-i} c_i$ for $\lambda > 1$, and suppose all secondary costs are bounded, $|c_i| \leq M$. Then there exists Λ such that for all $\lambda > \Lambda$,*

$$\mathbf{c} \preceq_{\text{lex}} \mathbf{c}' \iff C_\lambda(\mathbf{c}) \leq C_\lambda(\mathbf{c}').$$

In particular as $\lambda \rightarrow \infty$ the weighted objective C_λ induces exactly the lexicographic order, and the weight on the lawfulness coordinate c_0 diverges.

Proof. If $\mathbf{c} \prec_{\text{lex}} \mathbf{c}'$ they first differ at some index p with $c_p < c'_p$, so $c'_p - c_p \geq \epsilon$ for the (discrete or bounded) resolution $\epsilon > 0$. Then

$$C_\lambda(\mathbf{c}') - C_\lambda(\mathbf{c}) = \lambda^{k-p}(c'_p - c_p) + \sum_{i>p} \lambda^{k-i}(c'_i - c_i) \geq \lambda^{k-p}\epsilon - 2M \sum_{i>p} \lambda^{k-i}.$$

The positive term is order λ^{k-p} and the tail is $O(\lambda^{k-p-1})$, so the difference is positive once $\lambda > \Lambda := 1 + 2M(k)/\epsilon$. Ties ($\mathbf{c} = \mathbf{c}'$) map to equality. Hence the induced orders coincide for $\lambda > \Lambda$; letting $\lambda \rightarrow \infty$ the lawfulness coordinate's weight λ^k dominates all others, which is the statement that no finite secondary cost can purchase an unadmitted plan. $\square$

Theorem 4.2 is the precise sense of “you cannot buy your way past the first letter”: the price of unlawfulness is not high but infinite in the limit that defines the order.

Chapter 5

The Geometry of Conformance

5.1 The marking polytope

Token-flow execution — Petri nets, POWL, the lifecycle typestate — has a native geometry. Let there be p places; a *marking* is $\mathbf{m} \in \mathbb{Z}_{\geq 0}^p$, and a transition t is a pair $(\mathbf{m}_t^-, \mathbf{m}_t^+)$ of pre- and post-conditions. Firing t at $\mathbf{m}$ is defined iff $\mathbf{m} \geq \mathbf{m}_t^-$ (coordinatewise) and yields $\mathbf{m}' = \mathbf{m} - \mathbf{m}_t^- + \mathbf{m}_t^+$.

Definition 5.1 (Reachability cone and state equation). Stack the transition displacements $\delta_t = \mathbf{m}_t^+ - \mathbf{m}_t^-$ as columns of the *incidence matrix* $N \in \mathbb{Z}^{p \times |T|}$. If a firing sequence fires each transition t a total of $x_t \geq 0$ times (the *Parikh vector* $\mathbf{x}$), then any reachable marking satisfies the *state equation*

$$\mathbf{m} = \mathbf{m}_0 + N\mathbf{x}, \quad \mathbf{x} \in \mathbb{Z}_{\geq 0}^{|T|}. \quad (5.1)$$

The set of $\mathbf{m}$ satisfying (5.1) for some $\mathbf{x} \geq 0$ is the integer points of the translated cone $\mathbf{m}_0 + \text{cone}(N)$.

Proposition 5.1 (Conformance is membership; nonconformance is a separating hyperplane). *A trace with Parikh vector $\mathbf{x}$ is language-conformant only if its induced marking lies in the polytope $\mathcal{P} = \{\mathbf{m}_0 + N\mathbf{x} : \mathbf{x} \geq 0\} \cap \{\mathbf{m} \geq 0\}$. If an observed marking $\mathbf{m}^\dagger \notin \mathcal{P}$, then by the Farkas lemma there exists $\mathbf{y}$ with $\mathbf{y}^\top N \leq 0$ and $\mathbf{y}^\top (\mathbf{m}^\dagger - \mathbf{m}_0) > 0$: a certificate (separating hyperplane) of nonconformance.*

Proof. Membership in the finitely generated cone is the state equation with $\mathbf{x} \geq 0$. Farkas' lemma applied to the system $N\mathbf{x} = \mathbf{m}^\dagger - \mathbf{m}_0$, $\mathbf{x} \geq 0$ yields either a nonnegative solution (conformance) or a vector $\mathbf{y}$ with $\mathbf{y}^\top N \leq 0$ and $\mathbf{y}^\top (\mathbf{m}^\dagger - \mathbf{m}_0) > 0$ (the certificate). The two alternatives are mutually exclusive, which is exactly a decision with a proof. $\square$

Definition 5.2 (Conformance fitness as normalized distance). Let $\varphi(\sigma) = 1 - \frac{\text{tokens consumed on unenabled transitions}}{\text{tokens the replay attempted to consume}}$ on $[0, 1]$. Then $\varphi = 1$ iff the replay never attempted a disabled firing, i.e. the trace stays within $\mathcal{P}$ at every step; and $1 - \varphi$ is a normalized ℓ^1 measure of excursion outside the enabled set — a distance to the polytope boundary.

Theorem 5.2 (Lawful trace has unit fitness; the converse localizes the fault). *A trace is a firing sequence of the net iff $\varphi = 1$ under token replay. If $\varphi < 1$, the first frame at which a token is consumed on an unenabled transition is a coordinate-localized witness t^* of the violation.*

Proof. Token replay fires each event by removing pre-tokens and adding post-tokens; a firing is *enabled* iff $\mathbf{m} \geq \mathbf{m}_t^-$. If every event is enabled, no forced consumption occurs, numerator = 0, $\varphi = 1$, and by induction the sequence is a genuine firing sequence, so the marking stays in $\mathcal{P}$. Conversely if some event is not enabled, replay records a forced consumption at the earliest such index; that index is t^* , the numerator is positive, and $\varphi < 1$. $\square$

5.2 Separability: decomposing the geometry

The polytope of a realistic process is astronomically high dimensional, but the processes that occur in practice are *separable*: recursively built from sub-processes each of which is locally either pure choice (a state machine) or pure concurrency (a marked graph).

Theorem 5.3 (Separable decomposition, after Kourani–van der Aalst). *A sound workflow net that is separable admits a recursive decomposition into a POWL 2.0 model in which each node is either a choice graph (all exclusive/cyclic logic) or a partial order (all concurrent logic). Each node’s local marking geometry has dimension bounded by the node’s arity, independent of the global net size.*

Remark (Separability is a Rice quarantine for processes). Theorem 5.3 has an admission reading. *Separable* nets are exactly those whose geometry decomposes into bounded lanes; the decomposition algorithm terminates and certifies membership. Non-separable nets are *refused* with a stated reason. Thus separability is the admission predicate of the process layer: it is the decidable sub-language onto which arbitrary workflow logic is retracted, precisely as $\mathcal{O}^*$ retracts $\mathcal{O}$ in Chapter 2. The high-dimensional interior is real; each lane is bounded; the boundary artifact (the decomposition certificate) is small.

Chapter 6

The Instance: A Machine No One Held in His Head

6.1 Setting

We instantiate the theory on a concrete autonomous build. Over a bounded interval a fleet of automated agents performed exploration, planning, and implementation across a constellation of code repositories. The interior of this process consumed on the order of

$$\Sigma_{\text{tokens}} \approx 2.5 \times 10^6$$

machine-generated tokens — surveys, plans, diffs, tool transcripts — *no human read in full*. Let $\dim \Sigma$ denote the effective dimension of this interior; it is, for our purposes, unbounded relative to κ .

6.2 The Comprehension–Verification Gap

Corollary 6.1 (Comprehension–Verification Gap). *Let $\text{Comp}(\sigma)$ be the cost of comprehending the interior trace and $\text{Ver}(\sigma)$ the cost of verifying its boundary projection. Under Corollary 3.3 and Theorem 3.2,*

$$\text{Ver}(\sigma) = O(\kappa) \text{ per receipt}, \quad \text{Comp}(\sigma) = \Omega(\dim \Sigma),$$

so the ratio Comp/Ver grows without bound in $\dim \Sigma$ while verification stays constant. Trust is available at cost $O(\kappa)$ for a system of unbounded interior.

Proof. Verification reads only $\pi(\sigma)$, of size $\leq \kappa$, and performs a fixed hash recomputation and comparison per checked frame (Theorem 3.2, Corollary 3.3); hence $O(\kappa)$ per receipt. Comprehension requires ingesting the interior, whose size is $\Theta(\dim \Sigma)$; hence $\Omega(\dim \Sigma)$. The ratio therefore diverges. $\square$

6.3 Instantiation of the boundary

Concretely, the entire 2.5-million-token interior projects to a handful of κ -sized coordinates of the form

$$(\text{verdict} = \text{pass}, h_+ = \langle \text{256-bit hex} \rangle, \varphi = 1.0, \text{refusals} = k \text{ (each receipted)}),$$

together with a coverage figure $\text{pass_rate} \in [0, 1]$ over an enumerated frontier and a differential-oracle agreement flag. Each is at most a few chunks; each is checkable without reading the interior; and by Theorems 3.2 and 3.4 each is faithful to the interior it summarizes. The refusals are not omissions but recorded cells (Chapter 2, Proposition 2.2): a refusal register is the join-support of the denial monoid, materialized.

6.4 The antibody clause, formalized

Proposition 6.2 (Self-audited scale). *Let a system emit claims c and receipts $r(c)$. Define the overclaim set as claims lacking a faithful receipt: $\text{Over} = \{c : \nexists r(c) \text{ with } V(\pi) = 1\}$. If the system enforces $\text{Over} = \emptyset$ as an invariant (the *docs-exceed-mechanism* defect class), then the admissible magnitude of a claim is bounded solely by the mechanism that receipts its limits, not by the claim's rhetoric.*

Proof. By construction every admissible claim has a faithful receipt, verifiable at cost $O(\kappa)$ by Corollary 6.1; a claim without such a receipt is in Over and rejected by the invariant. Hence the set of admissible claims is exactly the set of receipted claims, whose magnitude is bounded by what the mechanism can receipt, not by what the narrative asserts. $\square$

This is the formal content of the doctrine: *a grand system without receipts is grandiosity; a grand system that receipts its own overclaims is machinery.*

Chapter 7

Conclusion

We set out to justify, rather than assume, the trust engineers already place in systems they cannot comprehend. The justification is a projection: admission retracts undecidable observation onto a decidable sub-language (Rice quarantine, Chapter 2); a deterministic bounded morphism manufactures artifacts from admitted observations (the Chatman equation, Chapter 3); and a collision-resistant chained receipt projects the unbounded execution trace onto a coordinate of size $\leq \kappa$ that a human can verify without reading the interior (the Faithful Projection Theorem). The calculus of the attention fluent explains scheduling and the infinite price of unlawfulness (Chapter 4); the geometry of the marking polytope makes conformance a membership question with separating-hyperplane certificates and localizes faults (Chapter 5); and separability is the process-layer admission predicate that keeps each lane bounded while the whole exceeds any head. The instance (Chapter 6) exhibits the theory operating on a 2.5-million-token interior whose trustworthiness rests entirely on boundary receipts.

The standard was never comprehension. The standard was a faithful projection, small enough to hold and guaranteed by the mechanism. Beyond cognitive load; within admissible projection.

Appendix A

Notation

Symbol	Meaning
$\mathcal{O}, \mathcal{O}^*$	observation space; admitted (decidable) sub-space
$\alpha, \perp$	admission retraction; refusal
$g_i, d(o)$	obligations; total denial word
$\mu, \mathcal{A}$	manufacturing morphism; artifact space
$\Sigma, \pi, \mathcal{R}$	execution traces; receipt projection; receipt space
$H, h_{\pm}$	collision-resistant hash; chain values
φ	conformance fitness / attention fluent
κ	human working-memory capacity (chunks), a fixed small constant
$N, \mathbf{m}, \mathbf{x}$	incidence matrix; marking; Parikh vector
$\preceq_{\text{lex}}$	lexicographic cost order